

DRUG TREATMENT OF ASTHMA

Learning Outcomes

Describe treatment of Bronchial Asthma

Outline Mechanism of action and adverse effects of drug used for treatment of Bronchial Asthma

BRONCHIAL ASTHMA

- **Asthma** is a disease characterized by an increased responsiveness of the trachea and bronchi to a variety of stimuli and manifests as narrowing of the airways that changes in severity either spontaneously or as a result of therapy.
- Inflammation underlying hyperreactivity.
- **Extrinsic asthma:** It is mostly episodic, and less prone to status asthmaticus.
- **Intrinsic asthma:** It tends to be perennial, status asthmaticus is more common.
- Avoid **NSAIDs, Aspirin in asthma** patients

The impairment of air flow in asthma is caused by three abnormalities:

- a. Constriction of bronchial smooth muscle (bronchoconstriction).
- b. Swelling of bronchiolar mucosa (bronchial edema).
- c. Excessive bronchial secretions.

Mast cells (present in lungs) and inflammatory cells recruited as a result of the initial reaction produce a multitude of mediators: histamine, $\text{TNF}\alpha$, PGs, LTs, PAF, interleukins, etc. which constrict bronchial smooth muscle, cause mucosal edema, hyperemia and produce viscid secretions, all resulting in reversible airway obstruction.

Drug Classes

I. Bronchodilators

A. Sympathomimetics: Salbutamol.

B. Methylxanthines: Theophylline.

C. Anticholinergics: Ipratropium bromide.

II. Leukotriene antagonists: Montelukast.

III. Mast cell stabilizers: Sodium cromoglycate.

IV. Corticosteroids

A. Systemic: Hydrocortisone.

B. Inhalational: Beclomethasone.

SYMPATHOMIMETICS

Examples: Adrenaline, salbutamol

- **Mechanism:** β 2-agonists are used in the symptomatic treatment of asthma.
- Drug of choice to relieve acute asthma exacerbation and prevent bronchoconstriction.
- **Side effects: Palpitation, restlessness, and nervousness**
- Should be cautiously used in hypertensive, heart patients and those receiving digitalis. (Digitalis is used in heart failure)
- Selectivity is further increased by inhaling the drug, so it is safe for inhalation.
- Inhaled salbutamol produces bronchodilatation within 5 min, and the action lasts for 2 to 4 hours.

THEOPHYLLINE

- **Mechanism:** These drugs inhibit the enzyme Phosphodiesterase and produce bronchodilatation, cardiac stimulation, and vasodilatation.
- Inhibition of PD enzyme → increases cAMP → Bronchodilatation
- It is well absorbed orally. It is distributed to all tissues and is 50% plasma protein-bound.
- **Side effects:** nausea, vomiting, headache, insomnia, tachypnea, epigastric pain, palpitation, hypotension, irritability.
- Interaction with ciprofloxacin, erythromycin causes an increase in theophylline levels in the blood and produces toxicity, i.e. arrhythmia.

Interactions

1. Agents that induce theophylline metabolism and decrease its plasma level are [smoking, phenytoin, rifampicin, and phenobarbitone](#).
2. Drugs that inhibit theophylline metabolism and increase its plasma level are—[erythromycin, ciprofloxacin](#), dose should be reduced to 2/3rd.
3. Theophylline enhances the effects of—[furosemide, hypoglycemics](#).

SODIUM CROMOGLYCAT

- **Mechanism:** Mast cell stabilizing effect.
- It inhibits the release of various asthma-provoking mediators e.g. histamine.
- It does not produce bronchodilatation
- Therefore not found beneficial in acute attacks of asthma and is used for prophylaxis only.

MONTELUKAST

- Indicated for the management of persistent asthma.
- **Mechanism:** Blockade of airway leukotriene receptors
- Useful alternative or adjunct to inhaled corticosteroid therapy.
- In severe asthma, they may permit a reduction in steroid dose.
- However, they are not to be used for terminating asthma episodes.

DRUGS THAT AFFECT THE ORAL CAVITY

Adverse drug effects

- An adverse drug reaction is defined by WHO as **‘a response to a drug which is noxious and unintended, and which occurs at doses normally used in man for the prophylaxis, diagnosis, therapy of diseases or for the modification of physiological function.**
- Since many patients take prescriptions and nonprescription medications, dentists should take a thorough medical history and be aware of medication-related problems and their potential effects on diagnosis and treatment planning

Salivary Gland Related Side effects

- **Dryness of mouth or Xerostomia**, results from diminished secretions of saliva. More than 250 medications claim xerostomia as a side effect.
- Few of the drugs are anticholinergics, antidepressants, anti-Parkinson's drugs, antihistamines, antipsychotics, diuretics, hypnotics, etc.
- Common oral manifestations resulting from decreased salivary flow include increased dental caries, fungal infections, bacterial infections, aphthous lesions, and dysphagia.
- **Pilocarpine and Bethanechol** have been suggested to be of potential use in the management of drug induced Xerostomia.

- **SALIVARY GLAND PAIN:** Antihypertensives, anti-thyroid agents, sulphonamides may cause salivary gland pain, as may drugs causing dry mouth.
- **DISCOLORATION OF SALIVA:** Discoloration of saliva and other body fluids into red or orange color may be seen in patients treated with Levodopa and rifampicin.
- **NONSPECIFIC ULCERATION:** Oral Ulceration may be caused as a result of direct application of over the counter drugs like Aspirin.

- **APHTHOUS ULCERATION:** Ulcers resembling recurrent aphthous stomatitis but have systemic causes are often termed aphthous-like ulcers.
- These include **barbiturates, beta-blockers, and NSAIDs.**
- **MUCOSITIS:** **Chemotherapy regimens** can cause Oral mucositis and ulceration.
- **BURNING MOUTH SYNDROME:** **ACE inhibitors, antibiotics.**
- **ERYTHEMA MULTIFORME (EM)** –syndrome consisting of symmetrical mucocutaneous lesions. Initial bullae may rupture, giving rise to widespread superficial ulceration, **Antibiotics, and NSAIDs.**
- **COLOR CHANGES OF ORAL MUCOSA AND TEETH:** **Anti-malarials and estrogen.**

DRUG INDUCED GINGIVAL ENLARGEMENT

- The drugs most commonly implicated are **Calcium channel blockers like nifedipine, antiepileptic drugs like Phenytoin, etc.**
- The gingival overgrowth is usually related to the dose of the drugs, the duration of therapy, the serum concentration, and the presence of dental plaque.
- Treatment is discontinuation of the offending drug, improvement of oral hygiene, and gingivectomy.

Swelling

- Around the mouth, the lips are the most frequently involved site, followed by the tongue.
- The swelling is acute and is often transient. Among the most common offending agents are, **penicillin, aspirin and other NSAIDs.**

Drug Class	Oral Effects	Examples
Antihypertensives	Gingival hyperplasia	Captopril, Nifedipine
Anticholinergics / Antidepressants	Xerostomia	Tricyclics, SSRIs
Chemotherapeutic agents	Mucositis, ulcers	Methotrexate, 5-FU
Antibiotics / Corticosteroids	Oral candidiasis	Doxycycline, Prednisone
Anticonvulsants	Gingival hyperplasia	Phenytoin